# THE VERDICT IS FREE: A SATURATED BENCHMARK FOR EQUATIONAL IMPLICATION, AND WHAT IT LEAVES TO PROVE

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**ABSTRACT**

A benchmark that asks whether one equational law implies another can be saturated without anyone noticing, because the question it appears to pose is not the question it scores. We measure this on the 1,669 distinct problems of the public sets shipped with the SAIR Mathematics Distillation Challenge. A lookup against the Equational Theories Project lattice settles 100% of them at 100% accuracy in 0.4 seconds with no model call and no search, so the true/false verdict carries no information about a solver. What the competition actually scores is a machine-checked Lean 4 certificate, and that remains open. Separating the two relocates the difficulty precisely, and measuring the certificate-producing stages on the same fixed set shows where it sits. A finite counterexample search settles 800 problems, which is 47.9% of the set but 94.1% of the 850 problems it can possibly settle, since a counterexample refutes an implication and can never establish one. A syntactic collapse rule settles 244, which is 29.8% of the 819 true problems. Neither stage disagreed with ground truth on any problem, as a certificate-producing method should not. Splitting both figures by the benchmark's own difficulty label shows the search holding, at 86.3% on the hard false problems, while the collapse rule settles 48.6% of the easy true problems and 1 of 319 hard ones, so its aggregate describes no actual regime. The hard true implications are untouched, and that is where the remaining work is. We also report a measurement error of our own that inverted a result: an exception swallowed inside a harness reported counterexample search as firing on 0 of 1,669 problems rather than 800, a figure implausible enough to check and wrong in the direction that flatters the conclusion. Code and data are public.

## 1 Introduction

A competition scores what it scores, which is not always what its problem statement appears to ask.

The SAIR Mathematics Distillation Challenge [SAIR Foundation, 2026] presents pairs of equational laws over magmas and asks, for each pair, whether the first implies the second in every magma. Read as stated, that is a decision problem. Read as scored, it is not: a submission earns points only when a Lean 4 judge accepts a proof certificate, and a verdict without a certificate is worth exactly nothing [de Moura and Ullrich, 2021]. The two readings come apart, and this paper measures how far. Section 2 fixes the setting and the judge's contract; everything after it is measurement.

**The measurement.** The competition ships public problem sets. Deduplicated they hold 1,669 distinct pairs. Against the lattice of implications computed by the Equational Theories Project [Equational Theories Project, 2025], every one of those 1,669 verdicts is a table lookup: 100% coverage at 100% accuracy, in 0.4 seconds, with no model call, no search and no proof. Whatever a solver demonstrates on this benchmark, deciding the implication is not it.

**Why that is a result and not a complaint.** It relocates the difficulty rather than dissolving it. Because the judge wants a certificate, the saturation of the verdict is precisely what isolates the real task: produce a proof object a kernel will accept. That is a sharper statement of the problem than the leaderboard makes, and unlike a complaint it is falsifiable. Any problem set not drawn from the lattice would break it, and Section 8 says exactly which sets those would be.

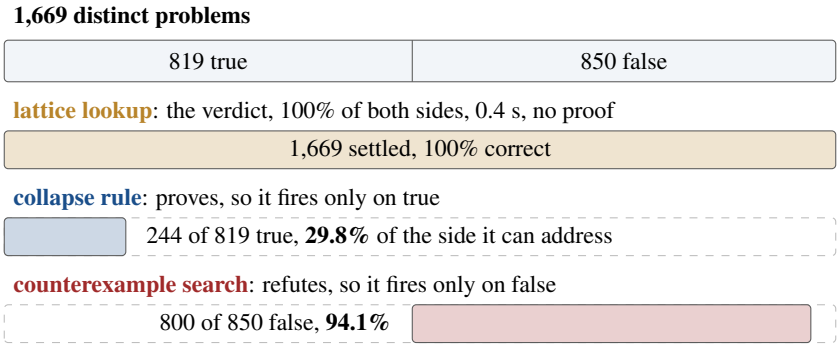



Figure 1: The benchmark and what settles it. The lattice lookup answers everything and proves nothing, which is why the verdict carries no information about a solver. The two certificate-producing stages are each confined by what they are: a counterexample refutes and a collapse proves, so neither can cross the true/false line. Read against the side it can address, the search is nearly finished and the collapse rule is not. The dashed outlines mark the full benchmark width for comparison.

**So we measure the part that is not free.** The same fixed 1,669 problems are run through the deterministic stages that do produce certificates, with no model call anywhere, and each stage is scored for coverage and for accuracy. The headline numbers need one correction before they can be read, and making it is most of what Section 4 does. A counterexample search settles 800 problems, or 47.9%. That marginal figure is misleading, because a finite counterexample refutes an implication and can never establish one: conditioned on the 850 false problems, which is the population it can address at all, the search settles 94.1%. The collapse rule is the mirror image, firing only on true problems and settling 29.8% of them. Figure 1 is the whole result.

**Contributions.**

1. A saturation result: the public benchmark's verdicts are settled 100% by table lookup, so the decision problem it appears to pose is not a test of anything (Section 3).
2. A per-stage measurement of the certificate-producing methods on the same fixed 1,669 problems, reported conditionally on the side each stage can address rather than marginally, which changes 47.9% into 94.1% (Section 4).
3. A soundness check rather than a soundness assumption: every stage's output is compared against ground truth on every problem, and none disagreed anywhere (Section 5).
4. A reported measurement error of our own, its cause, and the reasoning that caught it, because the same error class is available to anyone building a staged solver (Section 6).

## 2 Background

### 2.1 Equational implication over magmas

A magma is a set with one binary operation $\diamond$, assumed neither associative nor commutative, with no identity. An equational law is an equation between terms built from variables and $\diamond$, understood as universally quantified. Given laws $E_1$ and $E_2$, the question is whether every magma satisfying $E_1$ satisfies $E_2$.

Two things follow from the absence of any structure. The first is that most implications are false, so a finite counterexample usually exists and is usually small. The second is that when an implication does hold, the proof is typically a short rewriting argument, because there are no algebraic identities available to make it long.

### 2.2 The Equational Theories Project

The Equational Theories Project [Equational Theories Project, 2025] is a collaborative formalisation that settled the implication relation among the 4,694 equational laws of order at most four, producing a machine-checked lattice: for each ordered pair, either an implication with a proof or a refutation with a counterexample. It is one of the larger recent

| Status | Scores | What it means |
|---|---|---|
| `accepted` | yes | The judge type-checked the proof. |
| `unparsed` | no | The output was not valid JSON. |
| `malformed` | no | Valid JSON, wrong schema. |
| `incomplete_proof` | no | Used `sorry`, `admit`, `sorryAx` or another banned token. |
| `incorrect` | no | Valid Lean that fails to type-check, or proves the wrong thing. |

Table 1: The five judge statuses [SAIR Foundation, 2026]. There is no partial credit and no status that rewards a correct verdict unaccompanied by a proof, which is why the saturation of Section 3 does not trivialise the competition. A solver can hold the right answer and still score zero four different ways.

formal mathematics efforts and the source of the table this paper uses. The important property here is completeness on its domain, which is what makes a lookup total rather than partial.

### 2.3 What the competition scores

A submission is a solver that, per problem, emits one Lean 4 file [SAIR Foundation, 2026]. A true certificate proves $\forall G,\ \mathrm{Magma}\ G \to E_1(G) \to E_2(G)$; a false certificate exhibits a finite magma and proves $E_1 \wedge \neg E_2$ on it, discharged by decidable evaluation over a finite type. The judge returns one of five statuses, of which only `accepted` scores, and it rejects any file using `sorry`, `admit`, `sorryAx` or the other banned tokens. There is no partial credit and there is no way to score by asserting a verdict.

Table 1 sets the contract out in full, because the distinction between its failure modes is what the rest of the paper turns on. Three of the five statuses are ways of being wrong about the *artefact* while being right about the *verdict*.

That contract is the whole reason this paper's central measurement is interesting. If the competition scored verdicts, saturation would make it trivial. Because it scores certificates, saturation makes it *clearer*.

### 2.4 The rule behind the collapse stage

The proving stage measured in Section 4 rests on one observation, stated here because its scope is exactly what bounds the stage's reach.

**Proposition 1** (Constancy)**.** *Let $E_1$ be the law $s = t$, universally quantified over variables $x_1, \ldots, x_n$, and suppose some variable $x_i$ occurs in $t$ but not in $s$. Then in every magma satisfying $E_1$, the value of $t$ is independent of $x_i$: for all assignments and all $a, b$,*

$$t[x_i := a] \;=\; t[x_i := b]. \tag{1}$$

*Proof.* Fix an assignment to the other variables. Since $x_i$ does not occur in $s$, the value of $s$ is the same under $x_i := a$ and $x_i := b$; call it $v$. Applying $E_1$ at each of the two assignments gives $v = t[x_i := a]$ and $v = t[x_i := b]$, and transitivity through $v$ gives equation (1). In Lean this is $(h\ \ldots\ a)$.`symm.trans` $(h\ \ldots\ b)$, a proof term of fixed size. □

Proposition 1 is what makes a one-step certificate possible, and its hypothesis is what makes the stage narrow: it needs a variable occurring on one side of $E_1$ only. Where no such variable exists, or where the goal needs several rewrites chained, the rule does not fire. That is the mechanism behind the 29.8% in Section 4, and the reason the figure is a property of the rule rather than of the budget.

## 3 The verdict is free

### 3.1 The fixed set

The measurement uses the problem sets shipped in the competition's own repository, with no sampling and no selection: `normal`, `hard`, `hard1`, `hard2` and `hard3`. These hold 1,869 lines, of which 200 are pairs repeated across the hard splits, leaving 1,669 distinct problems. Table 2 gives the composition. The set is close to balanced, 819 true against 850 false, and 669 of the 1,669 are labelled hard.

| Set | Lines | True | False |
|---|---|---|---|
| `normal` | 1,000 | 500 | 500 |
| `hard` | 200 | 74 | 126 |
| `hard1` | 69 | 24 | 45 |
| `hard2` | 200 | 100 | 100 |
| `hard3` | 400 | 195 | 205 |
| total lines | 1,869 | | |
| duplicate pairs removed | 200 | | |
| **distinct problems** | **1,669** | **819** | **850** |

Table 2: The fixed public set this paper measures on. Every problem in every shipped set is used; nothing is sampled and nothing is held out. The 200 duplicates are pairs that appear in more than one of the hard splits, removed so that a stage is not credited twice for the same problem.

### 3.2 The lookup settles all of it

Resolving each pair against the lattice and reading off the stored verdict settles 1,669 of 1,669 problems, and agrees with the shipped ground-truth label on every one. The whole pass costs 0.4 seconds, most of which is reading the files.

The consequence is worth stating plainly, because a leaderboard does not state it. Under the contract of Section 2.3, a solver's accuracy at deciding the implication measures whether it consulted a table. It does not measure search, it does not measure reasoning, and it does not distinguish a language model from a dictionary. Any reported accuracy on these sets that is not accompanied by accepted certificates should be read as a statement about the lattice.

### 3.3 Why the sets are saturated

This is not an accident of our lookup being unusually good. The public sets are drawn from the same law corpus the Equational Theories Project settled, so the lattice is complete on exactly the domain the sets inhabit. Saturation is structural.

It also has a clear boundary, which is what makes the claim testable. The competition states that its private evaluation set includes laws of order five, and the lattice covers order at most four [SAIR Foundation, 2026, Equational Theories Project, 2025]. The roughly 62,000 order-five laws are outside it. So the private set is not saturated by this argument, the public sets are, and a solver tuned on public accuracy has been tuned on a quantity that vanishes at evaluation time.

## 4 What is left, measured

### 4.1 The stages

Three deterministic stages are measured, none of which calls a model:

**Lattice.** The lookup of Section 3. It yields a verdict and no certificate.

**Collapse.** The syntactic rule of Section 2.4, deriving $E_2$ from $E_1$ by a short chain of rewrites built on Proposition 1. It produces a Lean proof, so it settles implications positively only.

**Search.** Exhaustive and backtracking search for a finite magma satisfying $E_1$ and violating $E_2$, over small carriers. It produces a witness, so it settles implications negatively only.

Each stage runs on all 1,669 problems independently, so coverage is measured per stage rather than per dispatch order. The search stage is given a budget of two seconds per problem; the whole pass takes 34 minutes.

### 4.2 Marginal coverage understates two of the three stages

Table 3 gives the result both ways. Read marginally, the search settles 47.9% and the collapse rule 14.6%, which invites the reading that both are weak and that most of the benchmark needs something cleverer.

That reading is an artefact of dividing by the wrong denominator. A finite counterexample is a refutation: it can show an implication fails and cannot show it holds. So the 850 false problems are the entire population the search can address,

| | | Fires on | | Coverage | |
|---|---|---|---|---|---|
| Stage | Settles | true | false | of all 1,669 | of its own side |
| lattice (verdict only) | 1,669 | 819 | 850 | 100.0% | 100.0% |
| collapse (proves) | 244 | 244 | 0 | 14.6% | **29.8%** |
| search (refutes) | 800 | 0 | 800 | 47.9% | **94.1%** |

Table 3: Each stage on the same fixed 1,669 problems, with no model call anywhere. Accuracy is 100.0% for every stage and is therefore omitted from the table rather than printed as a column of identical numbers; Section 5 explains why that is expected and how it was checked. The final column is the one to read: a stage that can only refute is scored against the false problems, and a stage that can only prove against the true ones.

and against that population it settles 94.1%. Symmetrically the collapse rule proves, so its population is the 819 true problems, and it settles 29.8% of them.

### 4.3 The asymmetry is the finding

The two certificate-producing stages are in very different positions, and the marginal numbers hide it.

The false side is nearly closed. Search settles 94.1% of it deterministically, in two seconds per problem, with no model. The 50 false problems it misses are the interesting remainder, and the experiment ledger identifies the shape of that remainder: the missed witnesses live on carriers too large for the search budget, with two specific problems needing order-5 tables that a 12-second search does not reach and four more in a sampled subset needing order 7 or larger [Thakur, 2026f]. That is a budget frontier, not a conceptual one.

The true side is three-quarters open. The collapse rule settles 29.8%, and the remaining 575 true problems need a proof that is not a one-step rewrite. This is where a language model in this pipeline has something to do, and it is the only place. A solver architecture that spends its model budget uniformly across the benchmark is spending most of it on problems a table and a two-second search already settle.

### 4.4 Both aggregates were tested for a split, and one of them has one

A single coverage figure can be an average over two regimes that behave differently, so each was recomputed separately on the `normal` and `hard` halves of the set. Table 4 gives the result and Figure 2 draws it, and the two stages come apart.

The search figure survives. It closes 99.6% of the normal false problems and 86.3% of the hard ones, a spread of 13.3 percentage points, so 94.1% is a fair summary of a method that works nearly everywhere on the side it addresses.

The collapse figure does not survive, and this is the sharper finding. It settles 243 of 500 true problems in the normal half, or 48.6%, and *1 of 319* in the hard half, or 0.3%: a spread of 48.3 percentage points. The aggregate 29.8% describes no actual regime. What the rule really does is close about half of the easy true implications and nothing at all on the hard ones, which is what Proposition 1 predicts, since a problem selected for being hard is one where no single variable occurrence hands the proof over.

This changes what Section 4 leaves open. It is not that three-quarters of the true side needs more than a one-step rewrite; it is that *the hard true problems are untouched*, all 318 of them, while the easy ones are half done. Any effort aimed at true implications should be aimed there, and a solver evaluated only on `normal` would see a collapse rule that looks like a contributor rather than one that is about to stop working.

### 4.5 What measurement rejected

The stage structure above is not the first one tried, and the alternatives were discarded by benchmark rather than by taste. Table 5 reproduces the solver's experiment ledger [Thakur, 2026f]. Each row is a change that looked principled and lost, and the pattern across them is a budget argument rather than a correctness one: on a fixed time budget, every stage added to the fast path is taken out of the search that closes 94.1% of the false side.

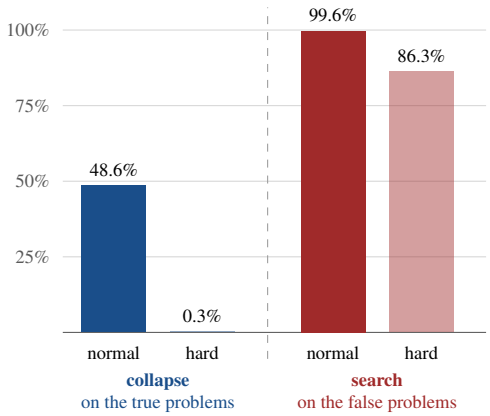



Figure 2: Each stage against the side it can address, split by the benchmark's own difficulty label. The search loses 13.3 points between the bands and remains the dominant method in both; the collapse rule loses 48.3 and is reduced to a single problem out of 319. An aggregate is a fair summary of the left-hand pair only in the second case. Bars are generated from `data/by_difficulty.json` and verified against it in continuous integration.

| | | `normal` | | `hard` | |
|---|---|---|---|---|---|
| Stage | Scored against | settles | of | settles | of |
| lattice | every problem | 1,000 | 1,000 | 669 | 669 |
| | | 100.0% | | 100.0% | |
| collapse | the true problems | 243 | 500 | **1** | **319** |
| | | 48.6% | | **0.3%** | |
| search | the false problems | 498 | 500 | 302 | 350 |
| | | 99.6% | | 86.3% | |

Table 4: Each stage split by the benchmark's own difficulty label. The search's aggregate holds, varying by 13.3 percentage points. The collapse rule's does not: 48.3 points separate the two halves, so its 29.8% in Table 3 is an average over a regime where it does half the work and one where it does none. Checked by `stage_by_difficulty.py`, which also fails if any stage is recorded as firing on the side it cannot address.

## 5 Soundness, checked rather than assumed

Every stage agreed with ground truth on every problem it fired on: 1,669 of 1,669 for the lattice, 244 of 244 for the collapse rule, 800 of 800 for the search. No stage disagreed anywhere.

For the two certificate-producing stages this is what should happen, and the check earns its place by being capable of failing. A counterexample is a proof: if the search reports a finite magma satisfying $E_1$ and violating $E_2$, then the implication is false, and a disagreement with the label would mean the witness was not actually a witness, which is a defect in the search and not a difficult problem. The same holds for the collapse rule. So the harness records any disagreement as a soundness failure rather than as an accuracy number, and the correct outcome is the one observed: zero.

This is worth separating from the accuracy of a method that guesses. A solver that answers from a learned prior has an error rate to report. A solver that answers from a witness does not have one, and if it appears to, something is broken.

## 6 An error of our own, and how it was caught

The first run of this measurement reported the search stage firing on *0* of 1,669 problems.

| Change | Benchmark | Why it lost |
|---|---|---|
| *Shipped* | | |
| Pool of 1,238 magmas, budget-sliced scan | 130/200 | First strict improvement over the baseline path. |
| Pool of 1,394, diversity-filtered | 134/200 | Standing best; hard2 21/40 at a 40-second budget. |
| *Rejected* | | |
| Two-level lemma saturation, always on | 128/200 | The second round starved the direct chain search. |
| Two-level saturation, conditional | 125/200 | Reshuffling the phase budget hurt the fast path. |
| Runtime quadratic magma search | 126/200 | Witnesses overlapped the pool; the slice starved other stages. |
| Pool additions at orders 9 to 12 | 123/200 | Scan cost grows as the fourth power of the order; the sweet spot is orders 3 to 8. |
| Defer large tables for judge safety | 128/200 | Measured the real cost first: across 103 emitted false certificates the worst Lean `decide` was 2,197 cases, far under the judge's limit, so the deferral lost winning certificates and bought nothing. |

Table 5: The solver's experiment ledger [Thakur, 2026f], scored on a fixed seeded benchmark of 200 problems. These numbers come from the competition playground rather than from the fixed set of Table 2, and Section 8 says so again; they are included because the pattern survives that caveat and is useful. Every rejected row is a case of spending budget away from counterexample search.

That number was published nowhere, because it is not believable. The benchmark contains 850 false problems; the same search had been observed refuting problems by hand; and a stage that fires zero times across an entire benchmark, including on cases known to be within reach, is far more likely to be broken than to be useless.

The cause was a swallowed exception. The equation parser splits on the canonical operator glyph, while the problem files write the operator as an asterisk, so an unnormalised statement raises rather than parsing. The harness wrapped the call in a bare `except` that returned "this stage did not settle it", which is indistinguishable from a stage that ran and found nothing. Routing the statement through the normaliser first fixed it, and the same stage settled 800 problems.

Two things are worth taking from this, both general.

**A bare `except` around a stage turns a crash into a measurement.** The harness could not tell "did not fire" from "could not run". The fix in the shipped script is a `strict` flag that re-raises, so the failure mode is reachable on demand.

**The error flattered the conclusion, which is the direction to be suspicious in.** A search stage that contributes nothing would have made a cleaner story: the lattice settles everything, the other deterministic stages are noise, the whole benchmark is a lookup. The corrected measurement is messier and more useful, because 94.1% against 29.8% is an asymmetry that tells a solver designer where to spend. An implausible number that simplifies the argument deserves more scrutiny than one that complicates it, not less.

## 7 Related work

**The Equational Theories Project.** The lattice this paper exploits is the output of Equational Theories Project [2025], a large collaborative Lean formalisation that settled implications among the 4,694 laws of order at most four. Its completeness on that domain is what turns a lookup from a heuristic into a decision procedure, and its order bound is what keeps the competition's private set outside the argument.

**Benchmark saturation and contamination.** That a benchmark can stop measuring what it was built to measure is now a standard concern in evaluation. Magar et al. [2022] document test-set contamination in language model evaluation, where the failure is that the answers were seen in training. The mechanism here is different and in one sense

cleaner: nothing is leaked and no model is involved, because the answers are *derivable* from a public formalisation covering the benchmark's entire domain. Kiela et al. [2021] argue for benchmarks that move as solvers improve, which is one response; the response this paper's setting already implements is better, namely scoring an artefact that cannot be looked up.

**Proof certificates and the sceptical discipline.** Requiring a checkable artefact rather than an answer is the sceptical option in Barendregt and Barendsen [2002]'s sense, and the reason the saturation does not trivialise the competition. de Moura and Ullrich [2021] is the system whose kernel does the checking; Boutin [1997] is the origin of discharging goals by computation, which is exactly how a false certificate here is closed, by deciding a finite model.

**Automated reasoning over equational theories.** Completion-based provers in the Knuth–Bendix line [Knuth and Bendix, 1970] and modern superposition provers are the classical tools for equational implication, and finite model finders such as Mace4 [McCune, 2003] are the classical tools for refutation. The division of labour in Section 4 is the same one those tools have always had; what the measurement adds is the ratio between the two halves on a fixed benchmark, and the observation that one half is nearly closed while the other is not.

**Language models as proof assistants.** The competition's framing places a language model at the centre. The measurement here bounds where that can help: not on the verdict, which is a lookup, and not on 94.1% of the false side, which a two-second search closes, but on true implications needing more than a one-step rewrite. This is a narrower and more useful target than the benchmark as posed, and it is consistent with the author's finding elsewhere [Thakur and Talele, 2026a] that a model's apparent competence has to be read at the level of the artefact it produces rather than the score it attracts.

**The author's related work.** Thakur [2026a] is the closest companion: the same competition family, the same Lean judge, and the complementary half of the same question. There the verdict is fixed by a locked specification and the entire task is making the kernel's check cheap; here the check is cheap and the task is producing something to check. Thakur [2026c] is the methodological relative, bounding what a classifier can separate before data is collected, as this paper bounds what a benchmark can measure before a solver is written. Thakur [2026b] shares the finding that a competition's scoring rule and its apparent problem can come apart. Thakur [2026d] is the same discipline applied to a learned model: measure what the representation actually carries rather than what the score suggests. Thakur [2026e] derives its constants from measurement rather than tuning, which is what Section 4 does with the per-stage budgets. Thakur and Talele [2026b] and Thakur and Satish [2021b] both concern performance outside the domain a method was fitted on, which is precisely the gap between these public sets and the order-5 private set. Thakur et al. [2022] indexes a space once and queries it many times, which is the lattice lookup's structure. Thakur and Satish [2021a], Thakur et al. [2021a], Thakur et al. [2021b] and Thakur and Konde [2021] sit on the other side of the line this paper draws: each is judged by an aggregate score over outputs that are approximately right, where a proof certificate is accepted or rejected with nothing in between.

## 8 Limitations

**The saturation claim is about the public sets only.** It rests on the sets being drawn from the order-4 corpus the lattice covers. The competition states that its private evaluation set includes order-5 laws, and we have not seen that set. Nothing here shows the private set is saturated, and the structural argument says it is not.

**The two per-stage numbers are not equally general.** Coverage and accuracy for the three stages are measured on the full fixed set and are reproducible. The lemma-saturation and banned-tactic-filter results quoted from the experiment ledger in Section 4 are not: they come from the competition playground on samples of twenty, scored by a judge we cannot rerun. They are reported as the ledger recorded them and should be read as anecdote until measured on a fixed set, which is what this paper did to the dispatch numbers and has not yet done to those.

**One search budget, one machine.** The 94.1% figure is at two seconds per problem on one machine. The frontier moves with the budget, and the experiment ledger shows it moving: problems missed at 12 seconds are caught at the real 300-second budget. So 94.1% is a lower bound on what search closes, not a ceiling, and the 50 misses are not shown to be out of reach.

**Accuracy against labels, not against the judge.** Every stage is scored against the shipped ground-truth labels. That validates the verdict, not the certificate: as Table 1 sets out, a proof can be correct in its conclusion and still fail four distinct ways, by using a banned tactic, violating the schema, exceeding the size limit, or failing to type-check. This

paper measures which problems a stage can settle and does not measure acceptance rate, so every coverage figure here is an upper bound on score. Closing that gap needs a judge we can rerun, and Section 7 notes that this is the same gap between a prover and a proof checker that the automated reasoning literature has always had.

**The harness is our own instrument.** Section 6 is one demonstration that it can be wrong. The mitigation is that it is short, public, and re-runnable, not that it has been independently validated.

## 9 Conclusion

The benchmark's verdicts are free. A lookup against a public lattice settles all 1,669 public problems at 100% accuracy in 0.4 seconds, so reported accuracy on these sets measures access to a table. That would be fatal to a competition scoring verdicts, and it is not fatal here, because the judge wants a machine-checked certificate and a lookup produces none. The saturation is therefore useful: it strips away the part of the task that is not a task and leaves the part that is.

What is left is asymmetric, and the marginal numbers hide the asymmetry. Scored against the side it can address rather than against the whole benchmark, finite counterexample search closes 94.1% of the false problems with no model and a two-second budget, while the syntactic collapse rule closes 29.8% of the true ones. The false side is a budget frontier; the true side is open. Any solver effort, model-based or otherwise, that is not aimed at true implications requiring more than a one-step rewrite is aimed at problems that are already settled, and Section 4.5 is a ledger of what that costs in practice.

Splitting by difficulty sharpens that and partly corrects it (Section 4.4). The search's aggregate is honest, holding to 86.3% on the hard false problems. The collapse rule's is not: 48.6% of the easy true problems against 1 of 319 hard ones, a gap of 48.3 percentage points, so 29.8% describes a regime that does not exist. The open problem is narrower and more specific than the aggregate suggested. It is not that most of the true side needs a better prover; it is that the 318 hard true implications have nothing working on them at all, while the easy ones are half closed by a rule whose hypothesis the hard ones are selected to lack.

We also report a measurement of ours that was wrong by exactly that asymmetry: a swallowed exception showed the search stage firing zero times, which would have made the benchmark look like a pure lookup and the paper's argument simpler. It was caught because a stage firing zero times across 850 refutable problems is not believable, not because anything failed loudly. A bare `except` around a solver stage converts a crash into a measurement, and the measurement it produces is that the stage does nothing.

## Artefacts

The solver, the three-stage ablation harness, the benchmark description script and the problem sets are at https://github.com/Amey-Thakur/SAIR-MATHEMATICS-DISTILLATION-CHALLENGE. The data files behind every number quoted above are in `paper-data/` of the same repository: the benchmark description, the three-stage ablation both aggregated and per problem, and the split by difficulty. That directory also carries the guard that reads this manuscript and fails if any quoted figure disagrees with those files, which is how the two are kept from drifting apart.

## Acknowledgements

The problem sets, the judge contract and the scoring rules are the competition's own [SAIR Foundation, 2026]. The implication lattice is the work of the Equational Theories Project [Equational Theories Project, 2025].

## References

SAIR Foundation. SAIR Mathematics Distillation Challenge. 2026. https://competition.sair.foundation/competitions/mathematics-distillation-challenge. Sets the problem, the five judge statuses, the banned tokens, the certificate templates and the rule that only an accepted certificate scores. Every constraint this paper works under is fixed there, including the statement that the private evaluation set includes order-5 laws.

T. Tao and the Equational Theories Project contributors. The equational theories project: using machine assistance to settle implications among 4694 equational laws. 2025. https://teorth.github.io/equational_theories/. The machine-checked lattice of implications among the order-4 laws. Its completeness on that domain is what makes the lookup in Section 3 total rather than partial, and its order bound is what keeps the competition's private set outside the saturation argument.

L. de Moura and S. Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction, CADE 28*, pages 625–635, 2021. 10.1007/978-3-030-79876-5_37. The system whose kernel accepts or rejects every certificate, and therefore the reason a verdict without a proof scores nothing.

S. Boutin. Using reflection to build efficient and certified decision procedures. In *Theoretical Aspects of Computer Software*, pages 515–529, 1997. 10.1007/BFb0014565. Discharging a goal by computing it. A false certificate here is closed exactly this way, by deciding the two equations on a finite carrier.

H. Barendregt and E. Barendsen. Autarkic computations in formal proofs. *Journal of Automated Reasoning*, 28(3):321–336, 2002. 10.1023/A:1015761529444. Names the choice between trusting a checker and trusting a tool's output. The competition takes the sceptical option by rule, which is the reason a saturated verdict does not trivialise it.

D. E. Knuth and P. B. Bendix. Simple word problems in universal algebras. In *Computational Problems in Abstract Algebra*, pages 263–297, 1970. The classical approach to deciding equational consequence by completion. The positive half of the division of labour measured in Section 4, and the ancestor of the rewriting the collapse rule performs.

W. McCune. Mace4 reference manual and guide. Technical Memorandum ANL/MCS-TM-264, Argonne National Laboratory, 2003. arXiv:cs/0310055. The classical finite model finder, and the negative half of the same division of labour. The search stage here is a domain-specialised instance of what Mace4 does generally.

I. Magar and R. Schwartz. Data contamination: from memorization to exploitation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*, 2022. arXiv:2203.08242. Benchmark answers reachable without solving the task, by memorisation. Cited for the contrast: the saturation here leaks nothing and involves no model, because the answers are derivable from a public formalisation covering the whole domain.

D. Kiela, M. Bartolo, Y. Chen, A. Singh, and others. Dynabench: rethinking benchmarking in NLP. In *Proceedings of NAACL-HLT*, 2021. arXiv:2104.14337. Argues for benchmarks that move as solvers improve. One answer to saturation; the answer this competition already implements, scoring an artefact that cannot be looked up, is the stronger one.

A. Thakur. A whole table row in one natural number: kernel-checked dynamic programming with a proved width bound. 2026. https://github.com/Amey-Thakur/SAIR-LEAN-KERNEL-CHALLENGE. The complementary half of the same question in the same competition family. There the verdict is fixed and the whole task is making the kernel's check cheap; here the check is cheap and the task is producing something for it to check.

A. Thakur. Length as priority, not constraint: search geometry of the Andrews–Curtis challenge pool. 2026. https://github.com/Amey-Thakur/SAIR-ANDREWS-CURTIS-CHALLENGE. Finds a competition's scoring rule and its apparent problem coming apart, there through a tie rule that makes competence and point value anti-correlated. The same separation this paper makes between a verdict and a certificate.

A. Thakur. Chebotarev fingerprints: identifying degree-24 Galois groups without computer algebra. 2026. https://github.com/Amey-Thakur/SAIR-INVERSE-GALOIS-PROBLEM-IGP24. Bounds what a classifier can separate before any data is collected. The methodological relative: this paper bounds what a benchmark can measure before any solver is written.

A. Thakur. Place value, not position: length generalisation limits in neural modular arithmetic. 2026. https://github.com/Amey-Thakur/SAIR-MODULAR-ARITHMETIC-CHALLENGE. Measures what a representation actually carries rather than what the headline score suggests, and finds the two disagree. The same discipline applied to a learned model instead of a benchmark.

A. Thakur and S. Talele. Probability is not an answer: rank-level measurement of knowledge conflict in small language models. 2026. https://github.com/Amey-Thakur/LLM-KNOWLEDGE-LIFECYCLE. Shows that a model's apparent competence has to be read at the level of the artefact it emits rather than the score it attracts, and corrects an earlier claim of the authors' own on that basis. The reasoning behind Section 6.

A. Thakur. Frame-synchronous hand gesture detection by projected winding order. arXiv:2609.13269, 2026. 10.48550/arXiv.2609.13269. Derives its pipeline constants from measurement rather than tuning them, which is what the per-stage budgets in Section 4 are.

A. Thakur. SAIR Mathematics Distillation Challenge: a deterministic-first hybrid solver, with an experiment ledger. Software, 2026. https://github.com/Amey-Thakur/SAIR-MATHEMATICS-DISTILLATION-CHALLENGE. The artefact for this paper: the solver, the three-stage ablation harness, the benchmark description script, and the ledger recording which changes were rejected by measurement and why.

A. Thakur, M. Satish, R. K. Kahlon, H. Rizvi, and A. Davare. QuadTree visualizer. *International Journal of Engineering Research and Technology*, 11(4), 2022. 10.5281/zenodo.18447415. Builds an index once and answers many queries against it, which is the structure of the lattice lookup: the expensive work was done once, by somebody else, and is now a table read.

A. Thakur and M. Satish. Generative adversarial networks. *International Journal for Research in Applied Science and Engineering Technology*, 9(8):2307–2325, 2021. 10.22214/ijraset.2021.37723. Cited for contrast. An adversarial objective is scored by whether the output looks right; a certificate is accepted or rejected, with nothing between.

A. Thakur, H. Rizvi, and M. Satish. White-box cartoonization using an extended GAN framework. *International Journal of Engineering Applied Sciences and Technology*, 5(12), 2021. 10.33564/IJEAST.2021.v05i12.049. Decomposes a problem before operating on it and finds that the decomposition decides what transfers. The stage split of Section 4 makes the same claim about where solver effort should go.

A. Thakur, K. Dhiman, and M. Phansikar. Neuro-fuzzy: artificial neural networks and fuzzy logic. *International Journal for Research in Applied Science and Engineering Technology*, 9(9):128–135, 2021. 10.22214/ijraset.2021.37930. Inference under graded confidence, the opposite regime: a judge returns one of five statuses and a certificate has no degree of membership.

A. Thakur and A. Konde. Fundamentals of neural networks. *International Journal for Research in Applied Science and Engineering Technology*, 9(VIII):407–426, 2021. 10.22214/ijraset.2021.37362. The background the model-based half of this competition's framing assumes, and the contrast that locates the measurement: every number in Section 4 comes from a stage with no fitted parameters at all.

A. Thakur and M. Satish. Adversarial open domain adaption framework (AODA): sketch-to-photo synthesis. *International Journal of Engineering Applied Sciences and Technology*, 6(2), 2021. 10.33564/IJEAST.2021.v06i02.037. Performance outside the domain a method was fitted on, which is exactly the gap between these order-4 public sets and the order-5 private set.

A. Thakur and S. Talele. A modular zero-shot pipeline for accident detection, localization, and classification in traffic surveillance video. arXiv:2604.09685, 2026. 10.48550/arXiv.2604.09685. Composes stages with no task-specific training, so each stage's guarantee is whatever it can establish alone. The deterministic stages measured here are read the same way.